Sales Performance & Product Profitability Analysis for a Bike Store

Tools: Power BI · Power Query · DAX · M · CSV
Data Modeling: Star/Snowflake hybrid
Data Visualization: Interactive dashboards, KPI, drillthrough, slicers, conditional formatting

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1. Project Overview

The goal of this project was to build an interactive Power BI report for analyzing sales performance, salesperson effectiveness, regional results, and product profitability.

The report was designed to support sales managers in monitoring sales targets, comparing salesperson and regional performance, and identifying products with high or low profitability.

The dashboard allows users to analyze results by salesperson, territory, product category, subcategory, and time period.

2. Business Questions

The analysis focuses on four main areas:

Sales Performance

- Which salespeople generate the highest sales and profit?
- Which salespeople have the lowest target achievement?
- How do salesperson results change over time?

Sales Targets

- How well are salespeople achieving their targets?
- Which employees meet or exceed their targets?
- Which employees may need additional support?

Regional Performance

- Which territories generate the highest sales and profit?
- How do regions compare with each other?
- Which regions have the highest potential for improvement?

Product Profitability

- Which products generate the highest sales and profit?
- Which products have low or negative margins?
- Does high sales volume also mean high profitability?
- Which products may require changes in pricing, discounts, or assortment?

3. Data

The data was provided as part of the DataHero course and was supplied in CSV files.

4. Data Preparation and Modeling

4.1 Data was cleaned and transformed in Power Query.

Main steps included:

- importing CSV files,
- creating a folder path parameter,
- cleaning missing and invalid values,
- removing unnecessary and technical columns,
- setting correct data types,
- handling conversion errors,
- merging tables,
- preparing fact and dimension tables,
- creating a calendar table,
- preparing sales target data.

4.2 Data Model

The final model uses a Star/Snowflake hybrid structure.

Fact tables:

- FACT SalesOrderHeader
- FACT SalesOrderDetail

Dimensions:

- DIM Customer
- DIM SalesPerson
- DIM Product
- DIM SalesTerritory

Other:

- Calendar
- ProductSubCategory
- Measures Table
- SalesPerson Share

4.3 DAX Measures

DAX measures were created to analyze: Total Sales, Total Cost, Gross Profit, Gross Margin, Total Orders, Total Quantity, Discount, Sales Target Achievement, Salesperson Share.

These measures were used across KPI cards, charts, tables, rankings, and interactive analysis.

4.4 Visualizations

The report includes interactive visualizations that enable the analysis of sales performance at the overall organizational level, as well as by region, salesperson, and product. The dashboard includes slicers that allow users to filter the data by year, quarter, month, territory, category, and subcategory. The dashboard was designed to allow users to move from a general overview of performance to a detailed analysis of an individual salesperson or product.

The report includes, among others:

- KPI cards,
- matrix,
- tables,
- bar charts,
- line charts,
- donut chart,
- gauge,
- scatter chart,
- conditional formatting,
- drillthrough,
- slicers,

- interactive filters,
- tooltips.

4.5 Results Analysis

The prepared report allows users to analyze:

- sales performance of individual salespeople,
- sales target achievement,
- performance of individual regions,
- changes in sales performance over time,
- individual salespeople's share of sales,
- product profitability,
- the relationship between sales and margin,
- products generating the highest profit,
- products with low profitability or generating losses.

5. Report Structure

1 — Sales Overview

Provides a high-level view of sales performance, salesperson results, target achievement, and regional performance. Identify top performers, underperforming employees, and differences between regions.

2 — Salesperson Details

Provides a detailed view of an individual salesperson using drillthrough. Monitor individual performance and identify areas for improvement.

3 — Product Performance

Focuses on product sales and profitability. Identify profitable products, low-margin products, and products that may require pricing, discount, or assortment decisions.

6. Key Insights

The report helps identify several important business patterns:

- High sales do not always mean high profitability.
- Some products can generate strong sales while having low or negative margins.
- Sales target achievement can highlight employees who perform strongly as well as those who may need additional support.
- Regional comparisons help identify areas with strong performance and areas with potential for improvement.

- Product profitability analysis can support decisions about pricing, discounts, promotions, and assortment.

7. Business Recommendations

The analysis can support decisions such as:

- reviewing discounts for high-sales, low-margin products,
- reassessing prices of products with low or negative margins,
- reviewing low-sales, low-profitability products for possible assortment changes,
- promoting products with high margins but lower sales,
- providing additional support or training to salespeople with low target achievement,
- using regional differences to identify growth opportunities.